

## Technology Stack

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|----------------------|---------------------------------------------------|
| <b>Date</b>          | 1 October 2026                                    |
| <b>Team ID</b>       | SWTID-2026-2533                                   |
| <b>Project Name</b>  | EduGenie Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 2 Marks                                           |

### Define Your Technology Stack:

EduGenie is a lightweight, AI-powered educational assistant that helps students and self-learners understand difficult topics. It answers academic questions, explains concepts in simple words, generates three-question quizzes, summarizes long passages and builds step-by-step learning paths from beginner to advanced level. The backend is an asynchronous FastAPI application and the frontend is a simple HTML, CSS and JavaScript page served with Jinja2, so it runs smoothly even on ordinary laptops.

To balance speed, cost and quality, EduGenie uses a hybrid AI approach: the Google Gemini API provides cloud intelligence, while the small LaMini-Flan-T5 model runs locally for concept explanations. The code is managed with Git and GitHub and deployed on Vercel. The application does not use a database because it stores no user data.

| S.No | Architecture Component / Layer                                     | Technology Chosen                                                                                   | Justification / Purpose                                                                                                                                                                                                       |
|------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side<br>(e.g., React, Angular, HTML/CSS)         | HTML, CSS, JavaScript, Jinja2 Templates                                                             | A lightweight, responsive single page that loads instantly with no heavy framework. Jinja2 serves the page from the backend, and plain JavaScript calls the API and shows results, including quiz feedback.                   |
| 2    | Backend / Server-Side<br>(e.g., Node.js, Python/Django, Java)      | Python 3.10+, FastAPI, Uvicorn                                                                      | Fast and asynchronous, with minimal boilerplate and automatic API documentation at /docs. It suits many concurrent AI requests, and Python gives direct access to the AI libraries.                                           |
| 3    | AI / ML Layer                                                      | Google Gemini (Google GenAI SDK);<br>LaMini-Flan-T5-783M with Hugging Face Transformers and PyTorch | Gemini handles question answering, quizzes, summaries and learning paths in the cloud. The small, instruction-tuned LaMini model explains concepts locally on an ordinary CPU, and Gemini is the fallback if it cannot load.  |
| 4    | Database / Data Storage<br>(e.g., MySQL, MongoDB, PostgreSQL)      | None (stateless application)                                                                        | The first release stores no user data, which keeps the app simple and private. Quiz answers are held in the browser while the page is open. A database such as SQLite or PostgreSQL can be added later for progress tracking. |
| 5    | Cloud / Hosting / Deployment<br>(e.g., AWS, Azure, Heroku, Docker) | Vercel (live deployment);<br>local machine with Uvicorn (development)                               | Vercel gives free, quick deployment of the web app straight from GitHub. The API key is stored in Vercel's environment settings. Locally the app runs with one command, <code>uvicorn main:app --reload</code> .              |
| 6    | Version Control & CI/CD<br>(e.g., GitHub, GitLab, Jenkins)         | Git and GitHub                                                                                      | Lets the five team members collaborate, keeps a full history of changes and triggers deployments.                                                                                                                             |

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|------|-----------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | Third-Party APIs / Other Tools<br>(e.g., Stripe, Twilio, Google Maps) | Google AI Studio (Gemini API key), python-dotenv, Visual Studio Code | Google AI Studio issues the Gemini API key. python-dotenv loads the key from a .env file, and VS Code with a Python virtual environment is the development setup. |